Wacyl Meddour

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Education

Massachusetts Institute of Technology 2023 – 2027 (expected)
S.B. in Mathematics; S.B. in Artificial Intelligence and Decision Making
GPA 5.0/5.0.

Research Experience

Quantizations of Poisson vertex algebras Feb. 2026 – Oct. 2026
Mentor: Ilya Dumanski. Faculty supervisor: Roman Bezrukavnikov.
Established a correspondence between sheaf chiral quantizations of affine Poisson schemes and $\hbar$-adic vertex algebras quantizing their jet functions.

Lattice field algebras and moduli of quiver representations Feb. 2025 – Aug. 2026
Mentor: Miguel Moreira. Faculty supervisor: Davesh Maulik.
Associated field algebras to lattices equipped with non-symmetric pairings and investigated their commutants. Applied the results to the homology of the moduli stack of complexes of quiver representations.

MIT UROP+ Summer 2026
Mentor: Yonghwan Kim. Faculty supervisor: Paul Seidel.
Constructed a mod-p analogue of the Hutchings–Lee invariant of 3-manifolds, with an application to orbit-period detection.

MIT Summer Program in Undergraduate Research (SPUR) Summer 2025
Mentor: Mary Stelow. Faculty supervisor: Roman Bezrukavnikov.
Joint project with Veronika Moroz on Khovanov homology classes attained by Seifert surface cobordisms in the 4-ball. Implemented SageMath code automating the computations.

Undergraduate Research Opportunities Program 2023 – 2024
Supervisor: Peter Kempthorne.
Studied stability of autoregressive forecasting models for time series undergoing regime changes.

Papers

[1] *Chiral quantizations of affine schemes.* In preparation, 2026, 7 pp.

[2] *Field algebras associated with non-symmetric lattices.* In preparation, 2026, 30 pp.

[3] *Circle-valued Morse theory and Frobenius monodromy.* MIT UROP+ report, 2026, 12 pp.

[4] *Computing surface cobordism maps in Khovanov homology.* With Veronika Moroz. MIT SPUR report, 2025. [pdf]

Talks

Computing surface cobordism maps in Khovanov homology, MIT SPUR final presentations. Aug. 2025

The Prime Number Theorem, MIT Directed Reading Program. Jan. 2024

Teaching

Teaching assistant, 18.950 Differential Geometry, MIT

Lecturer, number theory for computer scientists, MIT IAP (6.S096)

Mentor, 18.100B Real Analysis and 18.701 Algebra I, Undergraduate Math Association

Grader, 18.091, 18.700, 18.701, MIT Mathematics

Grader, 6.1200 Mathematics for CS and 6.1220 Algorithms, MIT EECS

Outreach

Instructor, MIT Global Teaching Labs, Italy; taught calculus to high-school students — Jan. 2025

Coach, Algerian Mathematical Olympiad national team — Summer 2024

Awards

International Mathematical Olympiad, Bronze Medal — 2022

Balkan Mathematical Olympiad, Silver Medal — 2023

William Lowell Putnam Competition, rank 172 — 2023

Coursework

Graduate

18.725	Algebraic Geometry I	Reading	Hodge Theory (with D. Maulik)
18.726	Algebraic Geometry II	18.217	Combinatorial Theory
18.705	Commutative Algebra	18.675	Theory of Probability
18.965	Geometry of Manifolds I	18.619	Discrete Prob. & Stochastic Processes
18.745	Lie Algebras	6.7960	Deep Learning

Undergraduate

18.100B	Real Analysis	Math 137	Algebraic Geometry (Harvard)
18.701	Algebra I	18.950	Differential Geometry
18.702	Algebra II	18.650	Fundamentals of Statistics
18.102	Functional Analysis	18.424	Seminar in Information Theory
18.112	Functions of a Complex Variable	6.4610	Natural Language Processing
18.901	Introduction to Topology	6.1220	Design and Analysis of Algorithms
18.904	Seminar in Topology	6.3900	Introduction to Machine Learning

Languages and Skills

Arabic (native), French (fluent), English (fluent). | Python, SageMath, Mathematica.